

LINGAM CHARAN

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Portfolio: lingamcharan-portfolio.vercel.app

EDUCATION

Bachelor of Technology (B.Tech.)

Electronics Engineering (VLSI Design & Technology)

NRI Institute of Technology (NRIIT) | Expected Graduation: May 2027

- CGPA: 7.55 / 10

Diploma in Electronics and Communication Engineering

Hindu College of Engineering and Technology | Duration: 2021 – 2024

- Percentage: 84.6%

TECHNICAL SKILLS

- Hardware Description Languages (HDLs):** Verilog HDL, VHDL
- RTL Design:** Finite State Machines (FSMs), Control Path Design, Clock/Reset Domains, RTL Coding Standards, Digital Logic Design
- Digital Architecture Concepts:** Computer Architecture Fundamentals, Memory Interfaces (FIFO, SRAM), On-Chip Bus Protocols (concepts)
- Programming Languages:** C (Basic)
- Simulation & EDA Tools:** Intel Quartus Prime, ModelSim, Xilinx ISE, EDA Playground
- Analog & Mixed-Signal:** Op-Amps, ADC/DAC (Academic coursework)
- Lab Equipment & Hardware Tools:** Oscilloscopes, Soldering (Diploma lab coursework)

VLSI PROJECTS

Synchronous FIFO Memory Controller

Verilog, Intel Quartus Prime, ModelSim

- Designed and implemented a synchronous FIFO buffer in Verilog with an 8-bit wide, 16-deep memory array.
- Developed the controller logic using separate read/write pointers and status flags (`full` , `empty`) computed based on an internal item counter.
- Simulated and verified read, write, and boundary conditions (full/empty write/read blocks) using ModelSim waveform analysis.

FILO (First-In-Last-Out) Stack

Verilog, ModelSim

- Designed an 8-bit wide, 16-deep Last-In-First-Out (LIFO/FILO) hardware stack in Verilog.
- Designed stack pointer controls and counter logic to track stack depth, incorporating `full` and `empty` flag logic.
- Modeled and simulated simultaneous read and write (push-pop) concurrency cases to verify data routing and flag behavior under synchronous clocking.

PROFESSIONAL EXPERIENCE & TRAINING

Silicon RTL Design Intern

Maven Silicon | June 2025 (45 Days)

- Completed intensive RTL design training focused on ASIC/SoC design flows, logic synthesis concepts, and hardware description languages.
- Practiced synchronous design methodologies, RTL coding guidelines, and functional verification techniques.

LEADERSHIP & COMMUNITY INITIATIVES

Founder

TapeoutX (VLSI Student Community) | *April 2026 – Present*

- Founded TapeoutX, a student-driven VLSI media brand and community with presence across Instagram (350+), Discord (220+), Telegram (150+), and LinkedIn (70+).
- Built and deployed the TapeoutX community website (tapeoutx.vercel.app) as a hub linking Discord, Instagram, YouTube, and LinkedIn.
- Research and script content on VLSI learning paths, digital logic fundamentals, and industry insights for the community's Instagram and YouTube channels.
- Organized an AMA and a two-session webinar in collaboration with PrimeMOS Research & Solutions to bridge the academic-industry gap.

Technical Content Creator

Instagram Channel (VLSI & Verilog) | *April 2026 – Present*

- Publish technical content, logic design tutorials, and Verilog coding guidelines on a personal educational Instagram channel, growing an audience of 3,000+ followers.

Technical Coordinator (Event Automation)

SARITAM 2K26 | *Feb 2026*

- Conceptualized and managed the 'Think & Pick' digital electronics competition.
- Developed and deployed a custom web application to automate the quiz workflow and digitize scoring, streamlining event operations and resource tracking.